Exploratory Data Analysis (EDA)

Rx Drug Marketing Mix Modeling (MMIX)

Business Context

Pharmaceutical firms compete within therapeutic classes, where multiple brands seek to influence prescriptions through two primary channels:

- **Detailing**: physician-targeted promotion
- **DTCA**: direct-to-consumer advertising

The commercial mechanism is:

> Marketing investment → influences physicians and patients → drives prescriptions → generates sales

Business Objective

The goal of this analysis is to understand how marketing investments translate into sales outcomes across brands and therapeutic classes, and to identify patterns that should guide model design and budget decisions.

Core Business Questions

1. Which brands dominate portfolio revenue, and how concentrated is the portfolio?
2. How has portfolio performance evolved over time?
3. How are detailing and Direct-to-Consumer Ads (DTCA) budgets allocated across brands and classes?
4. Which channel appears more efficient before formal modeling?
5. Is there evidence of diminishing returns to marketing spend?
6. Do classes and brands differ in baseline sales and marketing sensitivity?
7. Are there structurally different marketing regimes, especially around zero Direct-to-Consumer Ads (DTCA)?
8. What does the panel structure imply for modeling and validation?

Scope of Analysis

This review encompasses an 11-year historical dataset, capturing the following parameters:

- Observation Period: 2013–2024
- Therapeutic Classes: PPI (Proton Pump Inhibitors), SSRI (Antidepressants), Statin (Cholesterol)
- Portfolio Size: 22 distinct brands
- Key Metrics Evaluated:
  - Top-line Sales in Billions of Dollars
  - Physician Detailing Spend in Millions of Dollars
  - Direct-to-Consumer Advertising (DTCA) Spend ($ Millions)

In [179…

```
print("\n[EDA] Importing libraries...")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
pd.set_option("display.max_columns", 100)
pd.set_option("display.float_format", lambda x: f"{x:,.4f}")

print("[EDA] Libraries loaded successfully.")
```

```
[EDA] Importing libraries...
[EDA] Libraries loaded successfully.
```

In [180…

```
print("\n[EDA] Loading dataset and standardizing column names...")

df = pd.read_excel("data/MMM_Drug_Data.xlsx")
```

```python
df = df.rename(columns={
    "Class": "class",
    "Agent": "agent",
    "Year": "year",
    "Detailing": "detailing",
    "Direct-to-Consumer Advertising (DTCA)": "direct_to_consumer_ad",
    "Actual Sales": "actual_sales"
})

print("[EDA] Dataset loaded successfully.")
display(df.head())
```

```
[EDA] Loading dataset and standardizing column names...
[EDA] Dataset loaded successfully.
```

	class	agent	year	detailing	direct_to_consumer_ad	actual_sales
0	PPI	Prevacid	2013	43.8805	0.0000	2.9083
1	PPI	Prilosec	2013	41.6480	0.0000	14.7235
2	SSRI	Luvox	2013	22.3667	0.0000	0.9120
3	SSRI	Paxil	2013	57.4608	0.0000	11.2677
4	SSRI	Prozac	2013	84.5333	0.0000	20.2600

In [181…

```python
print("\n[EDA] Running initial data checks...")

print("Shape:", df.shape)

print("\nColumn types:")
display(df.dtypes.to_frame("dtype"))

print("\nMissing values:")
display(df.isna().sum().to_frame("missing_count"))

print("\nYears covered:", df["year"].min(), "to", df["year"].max())
print("Number of classes:", df["class"].nunique())
print("Number of agents:", df["agent"].nunique())

print("\n[Business Insight]")
print("The dataset covers 12 years, 22 brands, and 3 therapeutic classes with no missing values.")
print("This means we can analyze brands individually, compare across brands, and track changes over tim
```

```
[EDA] Running initial data checks...
Shape: (207, 6)

Column types:
```

	dtype
class	str
agent	str
year	int64
detailing	float64
direct_to_consumer_ad	float64
actual_sales	float64

```
Missing values:
```

	missing_count
class	0
agent	0
year	0
detailing	0
direct_to_consumer_ad	0
actual_sales	0

```
Years covered: 2013 to 2024
Number of classes: 3
Number of agents: 22

[Business Insight]
The dataset covers 12 years, 22 brands, and 3 therapeutic classes with no missing values.
This means we can analyze brands individually, compare across brands, and track changes over time.
```

In [182…

```
print("\n[EDA] Checking agent (brand) coverage across years...")

panel = pd.crosstab(df["agent"], df["year"])

n_agents = df["agent"].nunique()
n_years = df["year"].nunique()
expected_rows = n_agents * n_years
actual_rows = len(df)

print(f"Expected rows in a balanced panel: {expected_rows}")
print(f"Actual rows observed: {actual_rows}")
print(f"Missing agent-year combinations: {expected_rows - actual_rows}")

display(panel)

# Diagnostic only: checking if any agents (brands) appear in later years but not in earlier years.
# Time-based split is intentional: random splits leak future data into training.
# Final modeling split: Train 2013-2021, Validation 2022-2023, Test 2024.
test_agents = set(df[df["year"] >= 2023]["agent"].unique())
train_agents = set(df[df["year"] <= 2020]["agent"].unique())

print("\n[EDA] Checking if any agents (brands) appear in test years but not in training years...")
print(f"Brands in test period (2023-2024) with no data in training period (<=2020): {test_agents - trai
print(f"Brands in training period (<=2020) with no data in test period (2023-2024): {train_agents - tes

print("\n[Business Insight]")
print("Not all agents (brands) have data for every year. Some entered the market later than others.")
print("Out of 264 possible brand-year combinations (22 brands × 12 years), only 207 exist.")
print("The remaining 57 are missing because those agents (brands) had not yet entered the market in tho
print("This is expected in pharmaceutical markets where agents (brands) launch, grow, and decline at di
print("")
print("This has a direct impact on modeling: if we train on earlier years and test on later years,")
print("some agents (brands) in the test set may have no history in the training set.")
print("For example, Pexeva only appears from 2021 onward (4 records total), so any model trained on pre
print("has never seen Pexeva and cannot reliably predict its sales.")
print("During modeling, Pexeva should either be excluded from test evaluation or handled carefully to a
```

```
[EDA] Checking agent (brand) coverage across years...
Expected rows in a balanced panel: 264
Actual rows observed: 207
Missing agent-year combinations: 57
```

year	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
agent												
Aciphex	0	0	0	1	1	1	1	1	1	1	1	1
Altoprev	0	0	0	0	0	0	1	1	1	1	1	1
Celexa	0	0	1	1	1	1	1	1	1	1	1	1
Crestor	0	0	0	0	0	0	0	1	1	1	1	1
Lescol	1	1	1	1	1	1	1	1	1	1	1	1
Lescol XL	0	0	0	0	1	1	1	1	1	1	1	1
Lexapro	0	0	0	0	0	0	1	1	1	1	1	1
Lipitor	0	1	1	1	1	1	1	1	1	1	1	1
Luvox	1	1	1	1	1	1	1	1	1	1	1	1
Mevacor	1	1	1	1	1	1	1	1	1	1	1	1
Nexium	0	0	0	0	0	1	1	1	1	1	1	1
Paxil	1	1	1	1	1	1	1	1	1	1	1	1
Paxil CR	0	0	0	0	0	0	1	1	1	1	1	1
Pexeva	0	0	0	0	0	0	0	0	1	1	1	1
Pravachol	1	1	1	1	1	1	1	1	1	1	1	1
Prevacid	1	1	1	1	1	1	1	1	1	1	1	1
Prilosec	1	1	1	1	1	1	1	1	1	1	1	1
Protonix	0	0	0	0	1	1	1	1	1	1	1	1
Prozac	1	1	1	1	1	1	1	1	1	1	1	1
Prozac Weekly	0	0	0	0	0	1	1	1	1	1	1	1
Zocor	1	1	1	1	1	1	1	1	1	1	1	1
Zoloft	1	1	1	1	1	1	1	1	1	1	1	1

```
[EDA] Checking if any agents (brands) appear in test years but not in training years...
Brands in test period (2023-2024) with no data in training period (<=2020): {'  Pexeva'}
Brands in training period (<=2020) with no data in test period (2023-2024): None

[Business Insight]
Not all agents (brands) have data for every year. Some entered the market later than others.
Out of 264 possible brand-year combinations (22 brands × 12 years), only 207 exist.
The remaining 57 are missing because those agents (brands) had not yet entered the market in those year
s.
This is expected in pharmaceutical markets where agents (brands) launch, grow, and decline at different
times.

This has a direct impact on modeling: if we train on earlier years and test on later years,
some agents (brands) in the test set may have no history in the training set.
For example, Pexeva only appears from 2021 onward (4 records total), so any model trained on pre-2021 da
ta
has never seen Pexeva and cannot reliably predict its sales.
During modeling, Pexeva should either be excluded from test evaluation or handled carefully to avoid mis
leading error metrics.
```

In [183…

```
print("\n[EDA] Generating summary statistics for key commercial variables...")

summary = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].describe()

summary_readable = summary.copy()
summary_readable["actual_sales"] = summary["actual_sales"].apply(lambda x: f"${x:,.2f}B")
summary_readable["detailing"] = summary["detailing"].apply(lambda x: f"${x:,.2f}M")
summary_readable["direct_to_consumer_ad"] = summary["direct_to_consumer_ad"].apply(lambda x: f"${x:,.2f

summary_readable = summary_readable.reset_index().rename(columns={"index": "Metric"})
display(summary_readable)

print("\n[Business Insight]")
print("Sales range from near $0B to $63B, and marketing spend varies just as widely.")
```

```
print("This tells us agents (brands) operate at very different scales, so a few large brands could domi
print("Detailing and Direct-to-Consumer Ads (DTCA) budgets also vary significantly, meaning agents (bra
```

```
[EDA] Generating summary statistics for key commercial variables...
```

	Metric	actual_sales	detailing	direct_to_consumer_ad
0	count	$207.00B	$207.00M	$207.00M
1	mean	$11.67B	$56.03M	$26.32M
2	std	$12.97B	$54.92M	$49.30M
3	min	$0.00B	$0.00M	$0.00M
4	25%	$0.99B	$3.86M	$0.00M
5	50%	$7.53B	$44.97M	$0.00M
6	75%	$19.61B	$85.12M	$39.36M
7	max	$63.22B	$243.33M	$240.35M

```
[Business Insight]
Sales range from near $0B to $63B, and marketing spend varies just as widely.
This tells us agents (brands) operate at very different scales, so a few large brands could dominate any
analysis.
Detailing and Direct-to-Consumer Ads (DTCA) budgets also vary significantly, meaning agents (brands) fol
low different marketing strategies.
```

1. Revenue Concentration

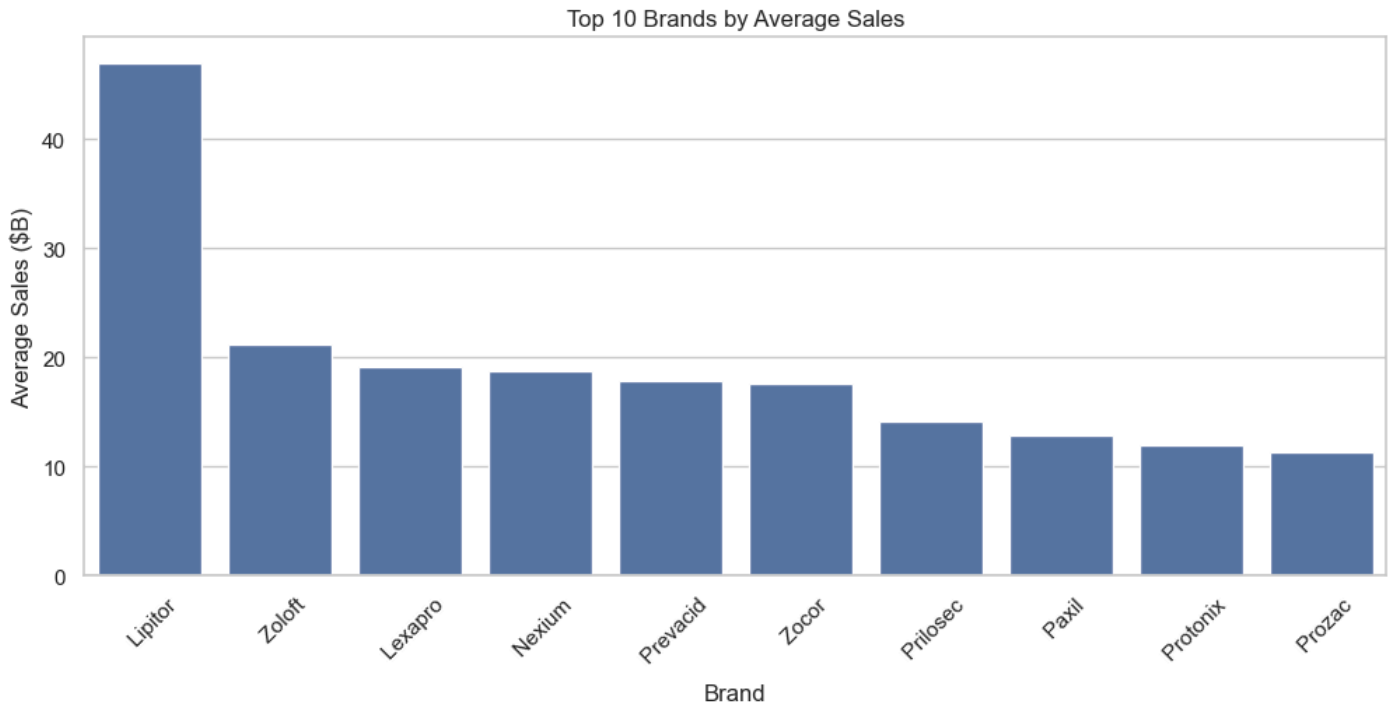
A first strategic question is whether sales are diversified or concentrated in a few blockbuster brands.

In [184…

```
print("\n[EDA] Ranking brands by average sales contribution...")

brand_sales = (
    df.groupby("agent", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

brand_sales["sales_share_pct"] = 100 * brand_sales["actual_sales"] / brand_sales["actual_sales"].sum()
brand_sales["cumulative_share_pct"] = brand_sales["sales_share_pct"].cumsum()

display(brand_sales.head(10))

plt.figure(figsize=(12, 5))
sns.barplot(data=brand_sales.head(10), x="agent", y="actual_sales")
plt.title("Top 10 Brands by Average Sales")
plt.xlabel("Brand")
plt.ylabel("Average Sales ($B)")
plt.xticks(rotation=45)
plt.show()

top_5_share = brand_sales.head(5)["sales_share_pct"].sum()

print("\n[Business Insight]")
print(f"The top 5 brands account for approximately {top_5_share:.1f}% of portfolio average sales.")
print("This means the portfolio is top-heavy. A few blockbuster agents (brands) drive most of the reven
print("so any model built on this data will be heavily influenced by how those top agents (brands) beha
```

```
[EDA] Ranking brands by average sales contribution...
```

	agent	actual_sales	sales_share_pct	cumulative_share_pct
7	Lipitor	47.0275	19.7929	19.7929
21	Zoloft	21.1441	8.8991	28.6920
6	Lexapro	19.1402	8.0557	36.7477
10	Nexium	18.7587	7.8951	44.6428
15	Prevacid	17.8747	7.5231	52.1659
20	Zocor	17.6050	7.4096	59.5754
16	Prilosec	14.1557	5.9578	65.5333
11	Paxil	12.8184	5.3950	70.9283
17	Protonix	12.0249	5.0610	75.9893
18	Prozac	11.2791	4.7471	80.7364




```
[Business Insight]
The top 5 brands account for approximately 52.2% of portfolio average sales.
This means the portfolio is top-heavy. A few blockbuster agents (brands) drive most of the revenue,
so any model built on this data will be heavily influenced by how those top agents (brands) behave.
```

2. Growth Trajectory

Before asking which marketing works, we need to understand whether the portfolio is expanding, maturing, or declining.

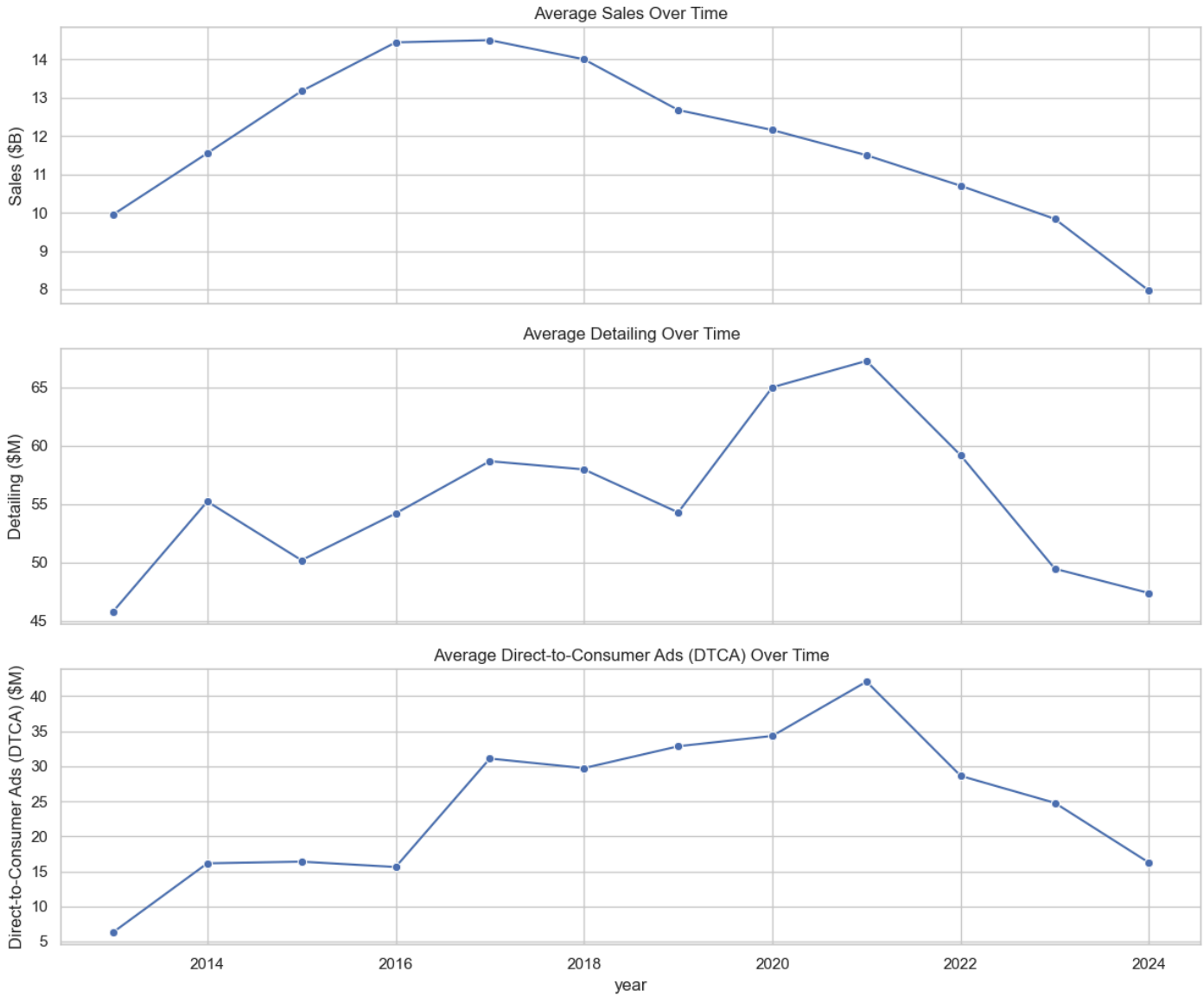
In [185…

```
print("\n[EDA] Analyzing portfolio growth trajectory over time...")

yearly = (
    df.groupby("year", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_ad"]]
    .mean()
)

yearly["sales_yoy_pct"] = yearly["actual_sales"].pct_change() * 100
yearly["detailing_yoy_pct"] = yearly["detailing"].pct_change() * 100
yearly["dtca_yoy_pct"] = yearly["direct_to_consumer_ad"].pct_change() * 100

display(yearly)

# Note: 'yearly' is already aggregated via .mean() above, so these are average values across brands per
fig, axes = plt.subplots(3, 1, figsize=(12, 10), sharex=True)

sns.lineplot(data=yearly, x="year", y="actual_sales", marker="o", ax=axes[0])
axes[0].set_title("Average Sales Over Time")
axes[0].set_ylabel("Sales ($B)")

sns.lineplot(data=yearly, x="year", y="detailing", marker="o", ax=axes[1])
axes[1].set_title("Average Detailing Over Time")
```

```
axes[1].set_ylabel("Detailing ($M)")

sns.lineplot(data=yearly, x="year", y="direct_to_consumer_ad", marker="o", ax=axes[2])
axes[2].set_title("Average Direct-to-Consumer Ads (DTCA) Over Time")
axes[2].set_ylabel("Direct-to-Consumer Ads (DTCA) ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Average sales grew from 2013 to around 2017, then steadily declined through 2024.")
print("This rise-and-fall pattern means we cannot assume a straight-line trend over time.")
print("The model will need to account for this non-linear trajectory, for example using year-level cont
```

```
[EDA] Analyzing portfolio growth trajectory over time...
```

	year	actual_sales	detailing	direct_to_consumer_ad	sales_yoy_pct	detailing_yoy_pct	dtca_yoy_pct
0	2013	9.9486	45.7758	6.3109	NaN	NaN	NaN
1	2014	11.5475	55.2130	16.1484	16.0714	20.6161	155.8812
2	2015	13.1646	50.1752	16.3910	14.0041	-9.1242	1.5025
3	2016	14.4348	54.1879	15.6212	9.6486	7.9974	-4.6969
4	2017	14.4936	58.6714	31.1021	0.4072	8.2740	99.1020
5	2018	13.9928	57.9632	29.7359	-3.4555	-1.2071	-4.3926
6	2019	12.6725	54.2737	32.8522	-9.4353	-6.3652	10.4801
7	2020	12.1514	65.0097	34.3327	-4.1124	19.7812	4.5066
8	2021	11.4936	67.2739	42.0747	-5.4132	3.4829	22.5497
9	2022	10.6978	59.1707	28.6292	-6.9237	-12.0451	-31.9561
10	2023	9.8339	49.4446	24.7726	-8.0753	-16.4374	-13.4710
11	2024	7.9631	47.3541	16.2398	-19.0238	-4.2278	-34.4443

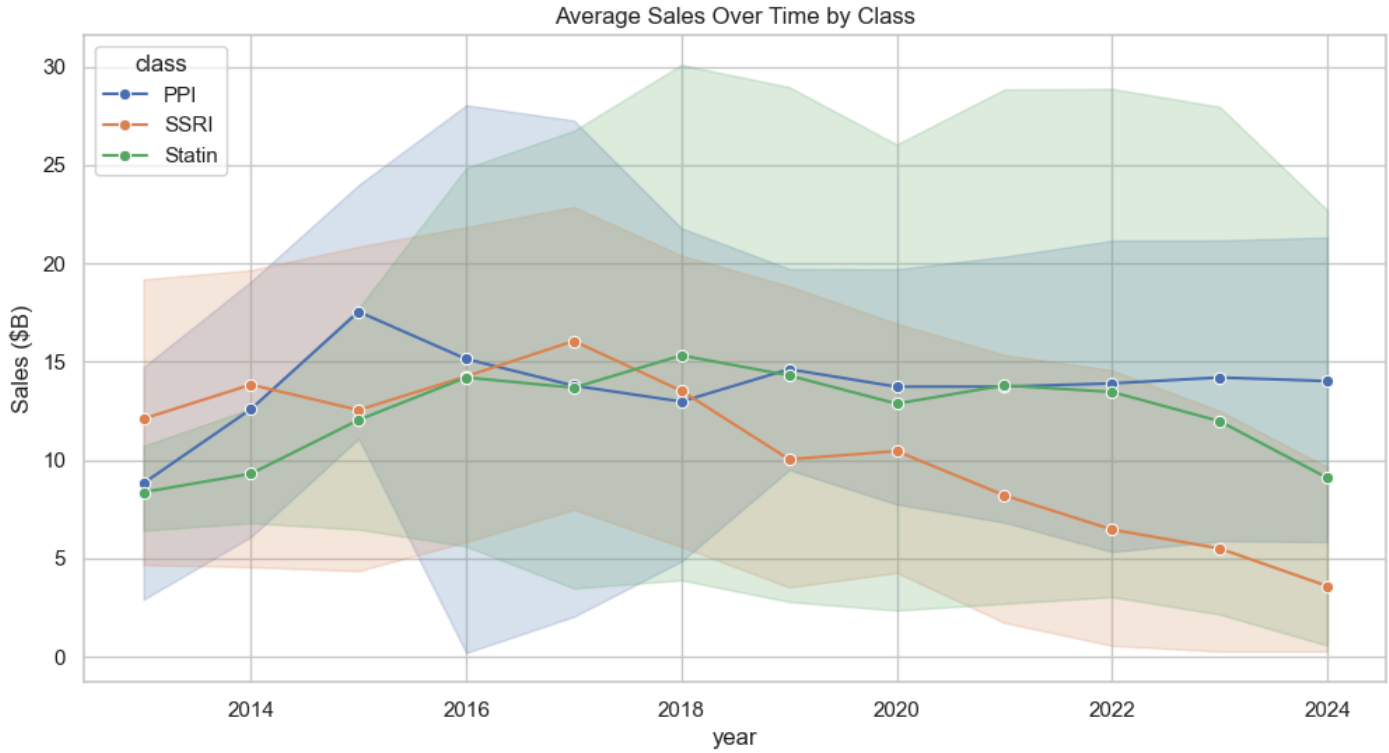



```
[Business Insight]
Average sales grew from 2013 to around 2017, then steadily declined through 2024.
This rise-and-fall pattern means we cannot assume a straight-line trend over time.
The model will need to account for this non-linear trajectory, for example using year-level controls.
```

In [186…

```
print("\n[EDA] Comparing class-level sales trends over time...")

plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x="year", y="actual_sales", hue="class", marker="o")
plt.title("Average Sales Over Time by Class")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("PPI, SSRI, and Statin classes each follow different sales trajectories over time.")
print("This means the therapeutic class a agent (brand) belongs to affects its sales pattern independen
print("The model should include class as a variable to capture these differences.")
```

```
[EDA] Comparing class-level sales trends over time...
```




```
[Business Insight]
PPI, SSRI, and Statin classes each follow different sales trajectories over time.
This means the therapeutic class a agent (brand) belongs to affects its sales pattern independently of m
arketing.
The model should include class as a variable to capture these differences.
```

3. Budget Allocation

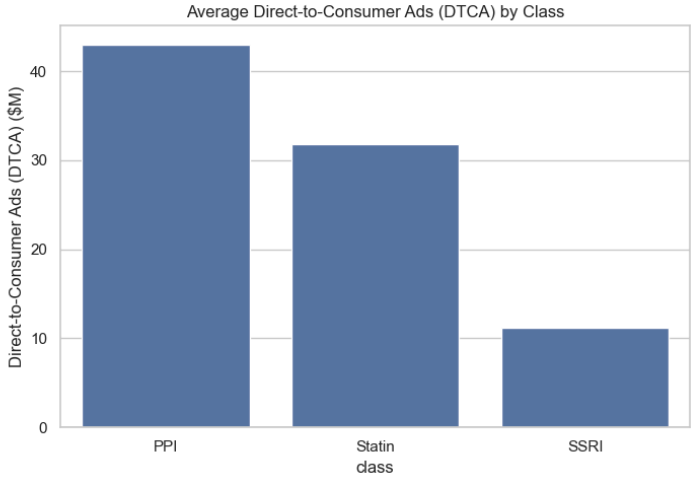
The next question is how the firm allocates detailing and Direct-to-Consumer Ads (DTCA) budgets across classes and brands.

In [187…

```
print("\n[EDA] Summarizing average budget allocation by class...")

class_spend = (
    df.groupby("class", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_ad"]]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(class_spend)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.barplot(data=class_spend, x="class", y="detailing", ax=axes[0])
axes[0].set_title("Average Detailing by Class")
axes[0].set_ylabel("Detailing ($M)")

sns.barplot(data=class_spend, x="class", y="direct_to_consumer_ad", ax=axes[1])
axes[1].set_title("Average Direct-to-Consumer Ads (DTCA) by Class")
axes[1].set_ylabel("Direct-to-Consumer Ads (DTCA) ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("PPI agents (brands) receive the most detailing and Direct-to-Consumer Ads (DTCA) spend, while SS
print("This means the company tailors its marketing channel mix based on therapeutic class.")
print("The model should allow for different marketing effects by class rather than assuming one univers
```

```
[EDA] Summarizing average budget allocation by class...
```

	class	actual_sales	detailing	direct_to_consumer_ad
0	PPI	13.8355	78.5221	42.9419
2	Statin	12.4964	48.5592	31.7618
1	SSRI	9.5923	49.9067	11.2310




```
[Business Insight]
PPI agents (brands) receive the most detailing and Direct-to-Consumer Ads (DTCA) spend, while SSRI agent
s (brands) receive the least Direct-to-Consumer Ads (DTCA).
This means the company tailors its marketing channel mix based on therapeutic class.
The model should allow for different marketing effects by class rather than assuming one universal respo
nse.
```

In [188…

```
print("\n[EDA] Identifying brands with the highest marketing investment...")

brand_spend = (
    df.groupby("agent", as_index=False)[["actual_sales", "detailing", "direct_to_consumer_ad"]]
    .mean()
    .sort_values("detailing", ascending=False)
)

display(brand_spend.head(10))

print("\n[Business Insight]")
print("Lexapro spends heavily on detailing but almost nothing on Direct-to-Consumer Ads (DTCA), while N
print("This brand-level variation in marketing strategy is important because the model needs enough div
print("in spending patterns to separately estimate the effect of each channel.")
```

```
[EDA] Identifying brands with the highest marketing investment...
```

	agent	actual_sales	detailing	direct_to_consumer_ad
6	Lexapro	19.1402	192.7008	0.7499
10	Nexium	18.7587	153.0828	182.8566
3	Crestor	7.9550	142.5837	119.2919
7	Lipitor	47.0275	124.0093	80.7884
15	Prevacid	17.8747	92.0651	38.7137
17	Protonix	12.0249	83.2735	3.3244
21	Zoloft	21.1441	80.4162	30.3664
20	Zocor	17.6050	70.3491	54.1879
0	Aciphex	5.8032	66.7499	1.1366
12	Paxil CR	4.8374	53.8639	24.0865

```
[Business Insight]
Lexapro spends heavily on detailing but almost nothing on Direct-to-Consumer Ads (DTCA), while Nexium in
vests heavily in both channels.
This brand-level variation in marketing strategy is important because the model needs enough diversity
in spending patterns to separately estimate the effect of each channel.
```

4. Baseline Efficiency

Before formal modeling, it is useful to compute simple descriptive efficiency ratios: how much revenue is associated with each unit of spend?

In [189…

```
print("\n[EDA] Computing descriptive baseline efficiency by channel...")

efficiency = df.copy()
efficiency["sales_per_detailing"] = np.where(
    efficiency["detailing"] > 0,
```

```
    efficiency["actual_sales"] / efficiency["detailing"],
    np.nan
)
efficiency["sales_per_dtca"] = np.where(
    efficiency["direct_to_consumer_ad"] > 0,
    efficiency["actual_sales"] / efficiency["direct_to_consumer_ad"],
    np.nan
)

efficiency_summary = (
    efficiency.groupby("class")[["sales_per_detailing", "sales_per_dtca"]]
    .mean()
    .sort_values("sales_per_detailing", ascending=False)
)

display(efficiency_summary)

avg_eff = efficiency[["sales_per_detailing", "sales_per_dtca"]].mean()
display(avg_eff.to_frame("portfolio_avg_efficiency"))

print("\n[Business Insight]")
print("Caution: These efficiency ratios are misleading for agents (brands) with very low or zero Direct
print("For example, SSRI shows extremely high Direct-to-Consumer Ads (DTCA) efficiency ($1,979 per $M),
print("agents (brands) with near-zero Direct-to-Consumer Ads (DTCA) spend, not by genuinely high return
print("These ratios should not be used for budget decisions. The parametric model will provide more rel
```

```
[EDA] Computing descriptive baseline efficiency by channel...
```

class	sales_per_detailing	sales_per_dtca
Statin	0.5474	59.6006
PPI	0.3913	335.6168
SSRI	0.3591	1,979.3762

	portfolio_avg_efficiency
sales_per_detailing	0.4387
sales_per_dtca	739.4620

```
[Business Insight]
Caution: These efficiency ratios are misleading for agents (brands) with very low or zero Direct-to-Cons
umer Ads (DTCA) spend.
For example, SSRI shows extremely high Direct-to-Consumer Ads (DTCA) efficiency ($1,979 per $M), but thi
s is driven by a few
agents (brands) with near-zero Direct-to-Consumer Ads (DTCA) spend, not by genuinely high returns.
These ratios should not be used for budget decisions. The parametric model will provide more reliable es
timates.
```

5. Distribution Shape and Market Structure

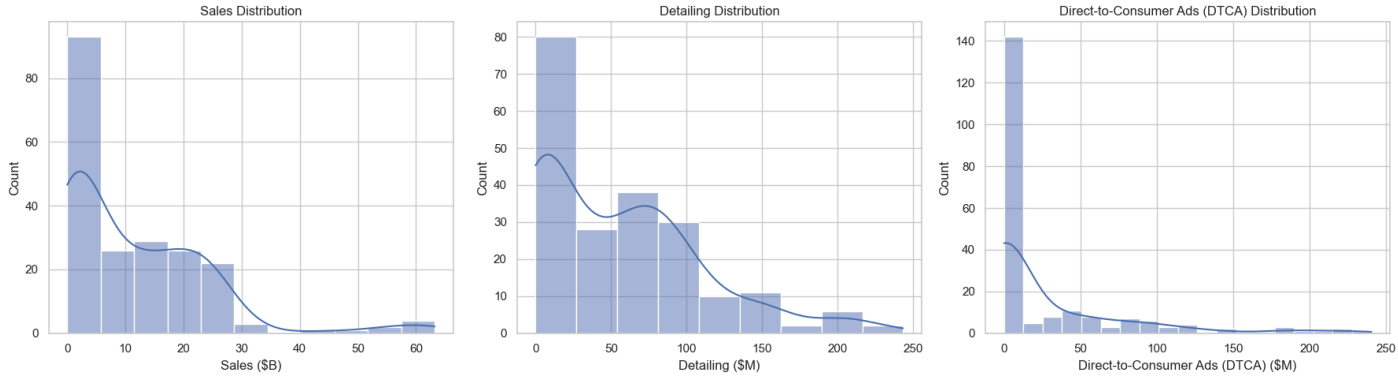
We now examine whether sales and spend are broadly distributed or concentrated in a small number of observations.

In [190…

```
print("\n[EDA] Visualizing raw distributions of sales and marketing spend...")

fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.histplot(df["actual_sales"], kde=True, ax=axes[0])
axes[0].set_title("Sales Distribution")
axes[0].set_xlabel("Sales ($B)")

sns.histplot(df["detailing"], kde=True, ax=axes[1])
axes[1].set_title("Detailing Distribution")
axes[1].set_xlabel("Detailing ($M)")

sns.histplot(df["direct_to_consumer_ad"], kde=True, ax=axes[2])
axes[2].set_title("Direct-to-Consumer Ads (DTCA) Distribution")
axes[2].set_xlabel("Direct-to-Consumer Ads (DTCA) ($M)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Sales, detailing, and Direct-to-Consumer Ads (DTCA) are all heavily skewed to the right.")
```

```
print("Most agents (brands) have low-to-moderate values, but a few (like Lipitor and Nexium) have very
print("This concentration means raw averages and simple regressions will be pulled toward these large a
```

```
[EDA] Visualizing raw distributions of sales and marketing spend...
```




```
[Business Insight]
Sales, detailing, and Direct-to-Consumer Ads (DTCA) are all heavily skewed to the right.
Most agents (brands) have low-to-moderate values, but a few (like Lipitor and Nexium) have very high val
ues.
This concentration means raw averages and simple regressions will be pulled toward these large agents (b
rands).
```

In [191…

```
print("\n[EDA] Quantifying skewness in raw variables...")

skewness_raw = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].skew()
display(skewness_raw.to_frame("skewness"))

print("\n[Business Insight]")
print("Skewness values above 1.0 confirm the distributions are significantly lopsided.")
print("Using raw values directly in regression would let a few extreme data points dominate the results
print("This is why we will apply log transformations before modeling.")
```

```
[EDA] Quantifying skewness in raw variables...
```

	skewness
actual_sales	1.7127
detailing	1.0358
direct_to_consumer_ad	2.2988

```
[Business Insight]
Skewness values above 1.0 confirm the distributions are significantly lopsided.
Using raw values directly in regression would let a few extreme data points dominate the results.
This is why we will apply log transformations before modeling.
```

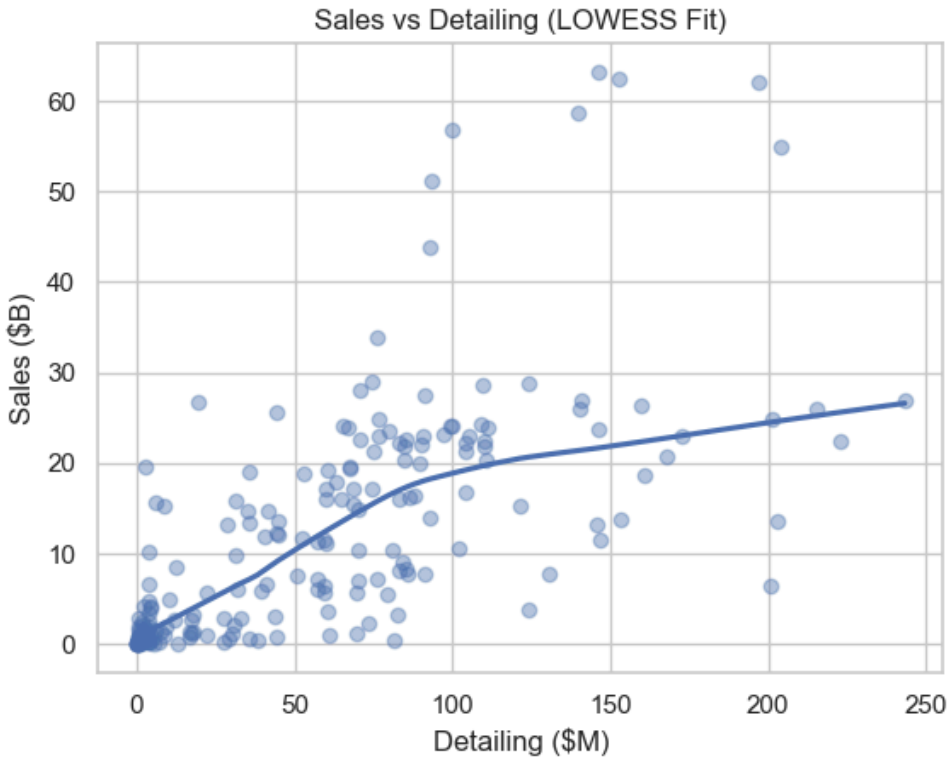
6. Spend Saturation and Diminishing Returns

The next business question is whether more spend always produces proportionally more sales.

In [192…

```
print("\n[EDA] Testing for diminishing returns in detailing using LOWESS...")

sns.regplot(
    data=df,
    x="detailing",
    y="actual_sales",
    lowess=True,
    scatter_kws={"alpha": 0.4}
)

plt.title("Sales vs Detailing (LOWESS Fit)")
plt.xlabel("Detailing ($M)")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("As detailing spend increases, the corresponding increase in sales gets smaller.")
print("For example, going from $0M to $50M in detailing adds more sales than going from $150M to $200M.
print("This diminishing returns pattern means doubling the budget will not double the sales impact.")
```

```
[EDA] Testing for diminishing returns in detailing using LOWESS...
```




```
[Business Insight]
As detailing spend increases, the corresponding increase in sales gets smaller.
For example, going from $0M to $50M in detailing adds more sales than going from $150M to $200M.
This diminishing returns pattern means doubling the budget will not double the sales impact.
```

In [193…

```
print("\n[EDA] Approximating where detailing saturation begins...")

df["detailing_bucket"] = pd.qcut(df["detailing"], q=10, duplicates="drop")
bucket_sales = df.groupby("detailing_bucket", observed=False)["actual_sales"].mean().to_frame("avg_sale

display(bucket_sales)

print("\n[Business Insight]")
print("Average sales rise from $0.37B in the lowest detailing bucket to $29.76B in the highest.")
print("But the gains slow down: the jump from bucket 7 to bucket 8 ($80-93M range) actually shows a sli
print("This confirms that additional detailing spend becomes less effective at higher levels.")
```

```
[EDA] Approximating where detailing saturation begins...
```

detailing_bucket	avg_sales
(-0.001, 0.639]	0.3719
(0.639, 2.96]	2.0414
(2.96, 6.907]	2.9686
(6.907, 28.631]	4.5371
(28.631, 44.969]	8.5018
(44.969, 67.258]	12.4589
(67.258, 80.046]	16.5847
(80.046, 92.864]	16.3139
(92.864, 134.211]	23.0129
(134.211, 243.33]	29.7583

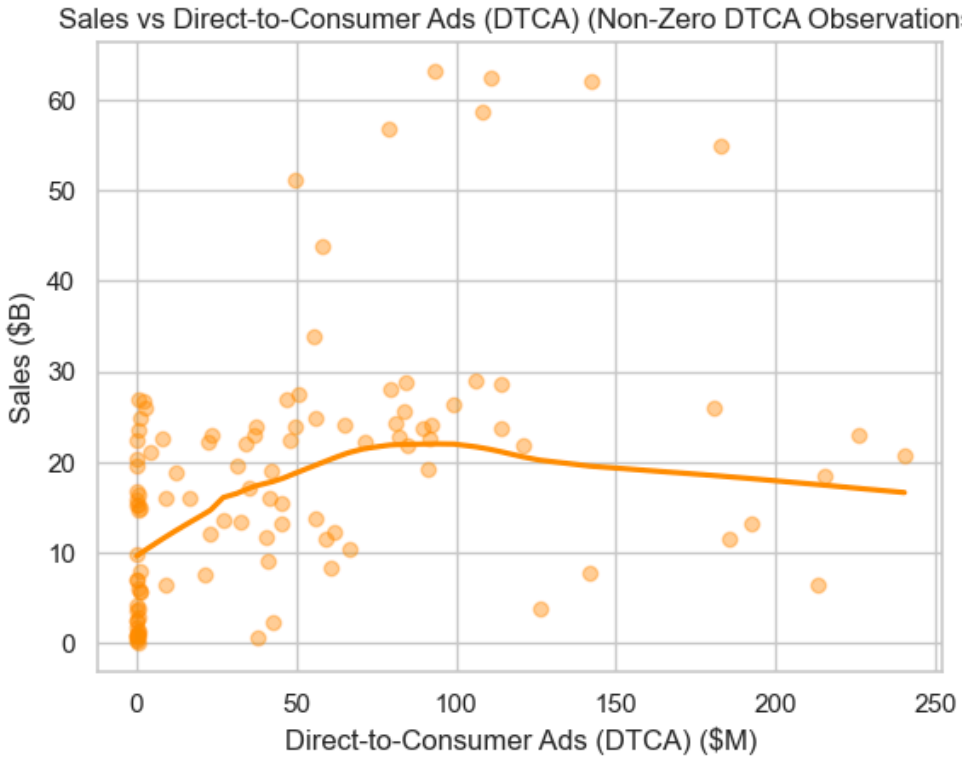
```
[Business Insight]
Average sales rise from $0.37B in the lowest detailing bucket to $29.76B in the highest.
But the gains slow down: the jump from bucket 7 to bucket 8 ($80-93M range) actually shows a slight decr
ease.
This confirms that additional detailing spend becomes less effective at higher levels.
```

In [194…

```
print("\n[EDA] Testing for diminishing returns in Direct-to-Consumer Ads (DTCA) among active advertiser

non_zero_dtca = df[df["direct_to_consumer_ad"] > 0]

sns.regplot(
```

```
    data=non_zero_dtca,
    x="direct_to_consumer_ad",
    y="actual_sales",
    lowess=True,
    scatter_kws={"alpha": 0.4},
    color="darkorange"
)

plt.title("Sales vs Direct-to-Consumer Ads (DTCA) (Non-Zero DTCA Observations)")
plt.xlabel("Direct-to-Consumer Ads (DTCA) ($M)")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("For agents (brands) that do invest in Direct-to-Consumer Ads (DTCA), higher spend is generally a
print("However, the relationship is not a straight line, and there is considerable scatter.")
print("This suggests Direct-to-Consumer Ads (DTCA) has a positive but variable effect, which the model
```

```
[EDA] Testing for diminishing returns in Direct-to-Consumer Ads (DTCA) among active advertisers...
```




```
[Business Insight]
For agents (brands) that do invest in Direct-to-Consumer Ads (DTCA), higher spend is generally associate
d with higher sales.
However, the relationship is not a straight line, and there is considerable scatter.
This suggests Direct-to-Consumer Ads (DTCA) has a positive but variable effect, which the model should c
apture using log or nonlinear terms.
```

7. Class and Brand Differences in Baseline Sales and Response

A key strategic issue is whether classes and brands differ structurally in both scale and likely marketing response.

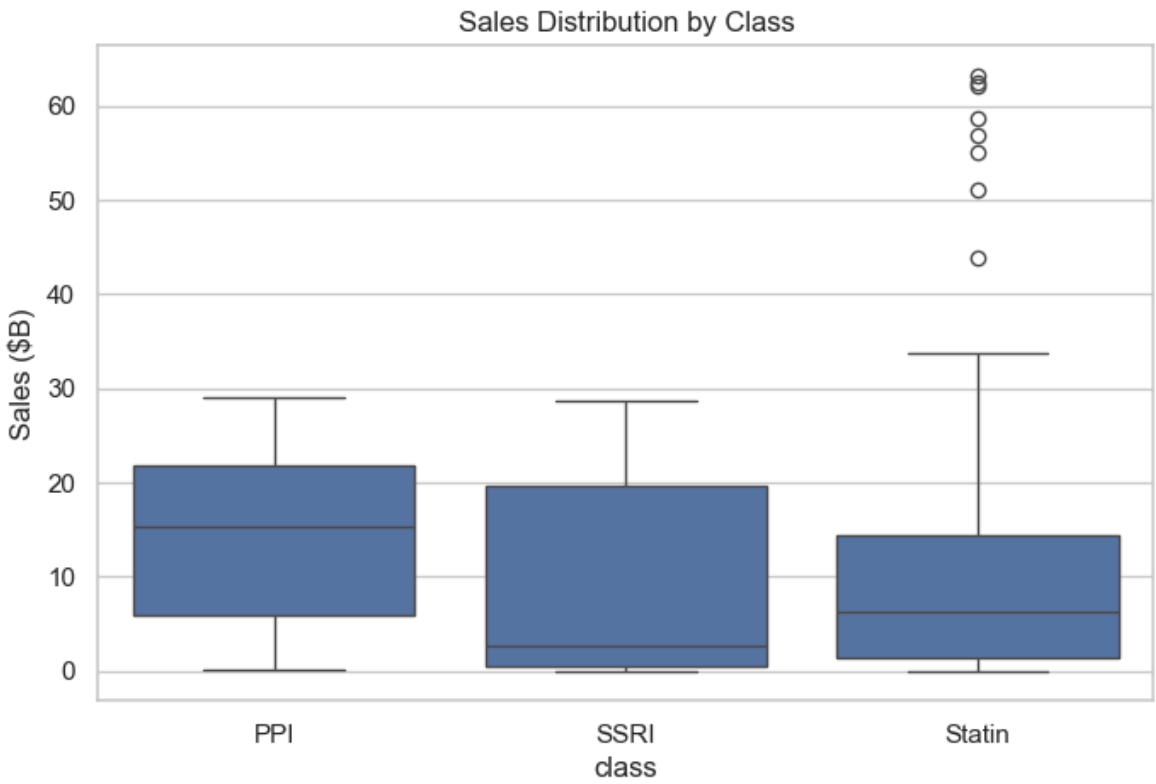
In [195…

```
print("\n[EDA] Comparing baseline sales by class...")

avg_sales_by_class = (
    df.groupby("class", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(avg_sales_by_class)

plt.figure(figsize=(8, 5))
sns.boxplot(data=df, x="class", y="actual_sales")
plt.title("Sales Distribution by Class")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("PPI agents (brands) average $13.8B in sales, Statins $12.5B, and SSRIs $9.6B.")
```

```
print("The spread within each class also differs: Statins have the widest range due to Lipitor.")
print("This means including class as a variable in the model will help explain baseline sales differenc
```

```
[EDA] Comparing baseline sales by class...
```

	class	actual_sales
0	PPI	13.8355
2	Statin	12.4964
1	SSRI	9.5923



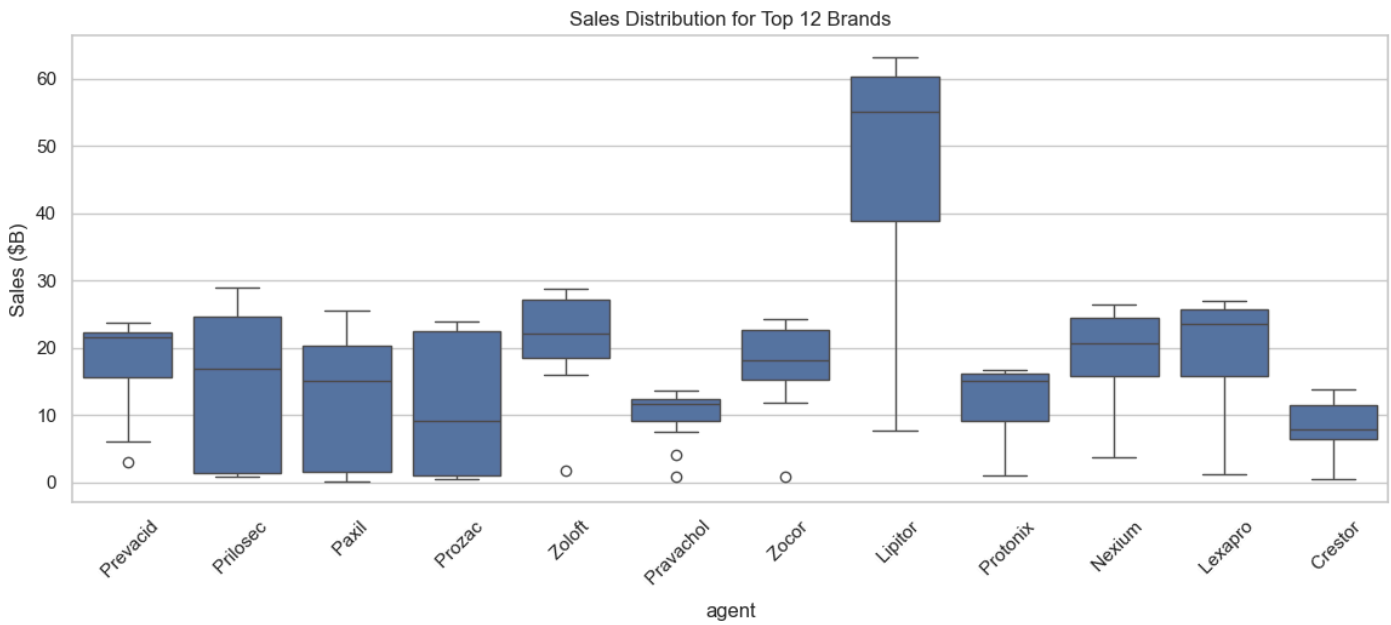

```
[Business Insight]
PPI agents (brands) average $13.8B in sales, Statins $12.5B, and SSRIs $9.6B.
The spread within each class also differs: Statins have the widest range due to Lipitor.
This means including class as a variable in the model will help explain baseline sales differences betwe
en agents (brands).
```

In [196…

```
print("\n[EDA] Comparing baseline sales by brand...")

avg_sales_by_agent = (
    df.groupby("agent", as_index=False)["actual_sales"]
    .mean()
    .sort_values("actual_sales", ascending=False)
)

display(avg_sales_by_agent.head(15))

top_agents = avg_sales_by_agent.head(12)["agent"].tolist()

plt.figure(figsize=(14, 5))
sns.boxplot(data=df[df["agent"].isin(top_agents)], x="agent", y="actual_sales")
plt.title("Sales Distribution for Top 12 Brands")
plt.ylabel("Sales ($B)")
plt.xticks(rotation=45)
plt.show()

print("\n[Business Insight]")
print("Lipitor averages $47B, while Luvox averages under $1B. These differences are too large to be exp
print("The model needs to include agent (brand) identity as a key variable to account for each brand's
```

```
[EDA] Comparing baseline sales by brand...
```

	agent	actual_sales
7	Lipitor	47.0275
21	Zoloft	21.1441
6	Lexapro	19.1402
10	Nexium	18.7587
15	Prevacid	17.8747
20	Zocor	17.6050
16	Prilosec	14.1557
11	Paxil	12.8184
17	Protonix	12.0249
18	Prozac	11.2791
14	Pravachol	9.9861
3	Crestor	7.9550
2	Celexa	7.8978
0	Aciphex	5.8032
12	Paxil CR	4.8374

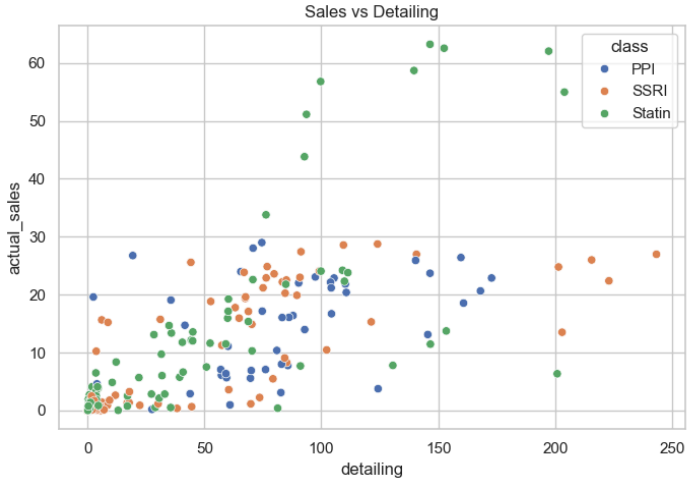


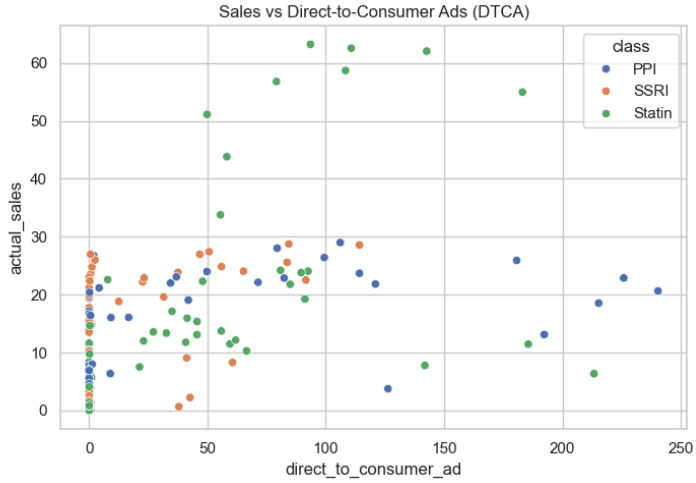

```
[Business Insight]
Lipitor averages $47B, while Luvox averages under $1B. These differences are too large to be explained b
y class alone.
The model needs to include agent (brand) identity as a key variable to account for each brand's inherent
market position.
```

In [197…

```
print("\n[EDA] Examining raw response relationships by class...")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.scatterplot(data=df, x="detailing", y="actual_sales", hue="class", ax=axes[0])
axes[0].set_title("Sales vs Detailing")

sns.scatterplot(data=df, x="direct_to_consumer_ad", y="actual_sales", hue="class", ax=axes[1])
axes[1].set_title("Sales vs Direct-to-Consumer Ads (DTCA)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("Plotting raw sales against raw marketing spend shows a lot of scatter with no clean pattern.")
print("This noise is partly because agents (brands) of very different sizes are mixed together.")
print("Log transformations will reduce this noise and make the underlying relationships easier to detec
```

```
[EDA] Examining raw response relationships by class...
```






```
[Business Insight]
Plotting raw sales against raw marketing spend shows a lot of scatter with no clean pattern.
This noise is partly because agents (brands) of very different sizes are mixed together.
Log transformations will reduce this noise and make the underlying relationships easier to detect.
```

8. Structural Zero-Direct-to-Consumer Ads (DTCA) Regime

A large share of observations show exactly zero Direct-to-Consumer Ads (DTCA), which may reflect a distinct strategic regime rather than just low spend.

In [198…

```
print("\n[EDA] Measuring the prevalence and structure of zero Direct-to-Consumer Ads (DTCA)...")

df["dtca_zero"] = (df["direct_to_consumer_ad"] == 0).astype(int)

zero_dtca = df["dtca_zero"].sum()
total = len(df)

print(f"Zero Direct-to-Consumer Ads (DTCA) observations: {zero_dtca}")
print(f"Total observations: {total}")
print(f"Share of zero-Direct-to-Consumer Ads (DTCA) observations: {zero_dtca / total:.2%}")

print("\nZero Direct-to-Consumer Ads (DTCA) by class:")
display(pd.crosstab(df["class"], df["dtca_zero"]))

print("\nZero Direct-to-Consumer Ads (DTCA) by year:")
display(pd.crosstab(df["year"], df["dtca_zero"]))

print("\n[Business Insight]")
print("Almost half (47%) of all observations have zero Direct-to-Consumer Ads (DTCA) spend.")
print("SSRI agents (brands) are most likely to skip Direct-to-Consumer Ads (DTCA) (47 out of 81 observa
print("This split suggests two distinct marketing strategies: agents (brands) that advertise to consume
print("The model should treat these as separate regimes, for example by including a zero-Direct-to-Cons
```

```
[EDA] Measuring the prevalence and structure of zero Direct-to-Consumer Ads (DTCA)...
Zero Direct-to-Consumer Ads (DTCA) observations: 98
Total observations: 207
Share of zero-Direct-to-Consumer Ads (DTCA) observations: 47.34%

Zero Direct-to-Consumer Ads (DTCA) by class:
```

dtca_zero	0	1
class		
PPI	32	16
SSRI	34	47
Statin	43	35

```
Zero Direct-to-Consumer Ads (DTCA) by year:
```

dtca_zero	0	1
year		
2013	3	7
2014	5	6
2015	6	6
2016	6	7
2017	7	8
2018	12	5
2019	12	8
2020	10	11
2021	13	9
2022	13	9
2023	10	12
2024	12	10

```
[Business Insight]
Almost half (47%) of all observations have zero Direct-to-Consumer Ads (DTCA) spend.
SSRI agents (brands) are most likely to skip Direct-to-Consumer Ads (DTCA) (47 out of 81 observations),
relying on physician detailing instead.
This split suggests two distinct marketing strategies: agents (brands) that advertise to consumers and a
gents (brands) that do not.
The model should treat these as separate regimes, for example by including a zero-Direct-to-Consumer Ads
(DTCA) indicator variable.
```

9. Log Transformations for Parametric Modeling

Because the raw variables are highly skewed and the commercial response appears multiplicative, we prepare log variables for the next stage.

In [199…

```
print("\n[EDA] Creating log-transformed variables for next-stage modeling...")

df["log_sales"] = np.log1p(df["actual_sales"])
df["log_detailing"] = np.log1p(df["detailing"])
df["log_dtca"] = np.log1p(df["direct_to_consumer_ad"])

print("[EDA] Log variables created: log_sales, log_detailing, log_dtca")
```

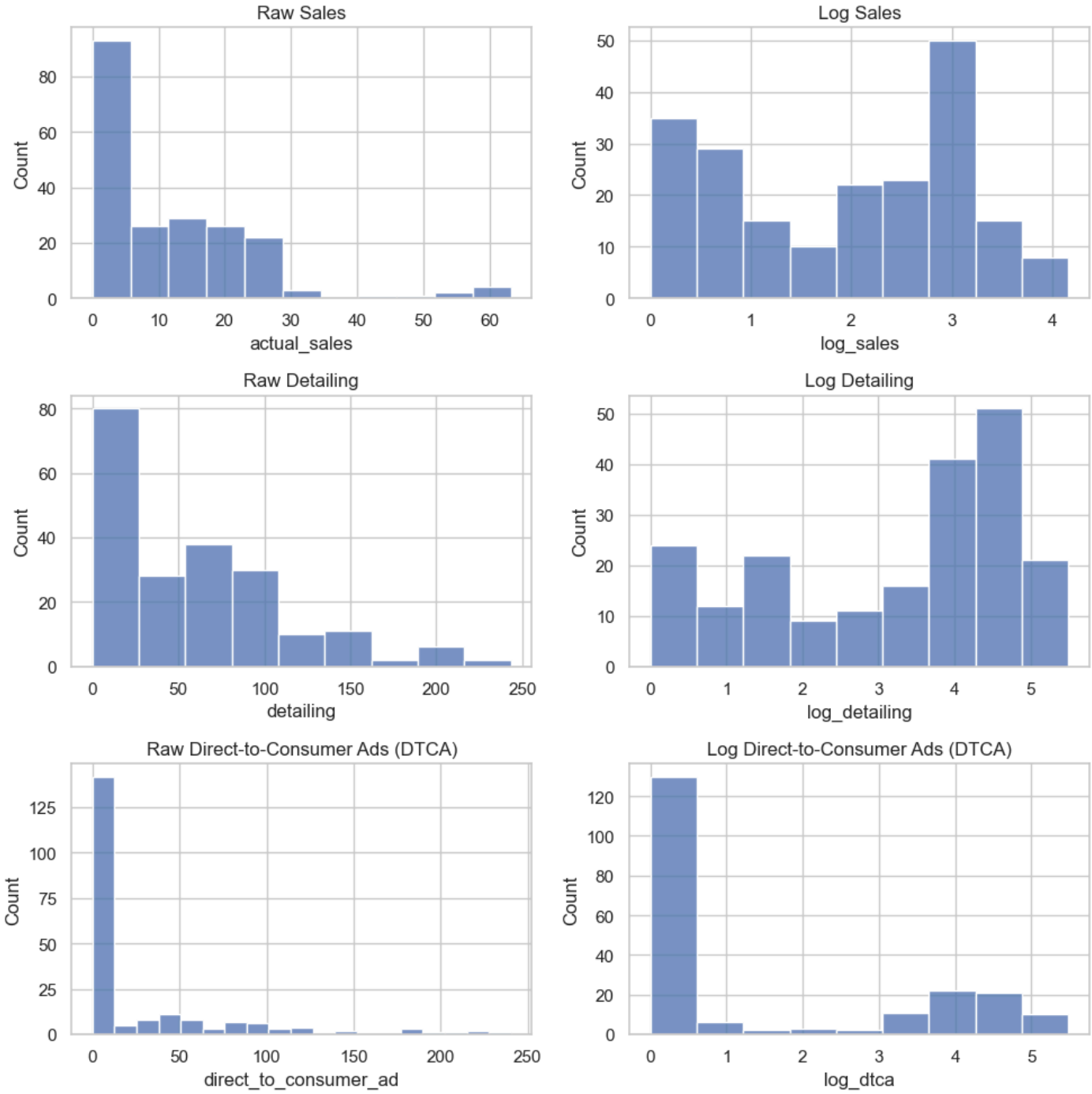
```
[EDA] Creating log-transformed variables for next-stage modeling...
[EDA] Log variables created: log_sales, log_detailing, log_dtca
```

In [200…

```
print("\n[EDA] Comparing raw and log-transformed distributions...")

fig, axes = plt.subplots(3, 2, figsize=(10, 10))

sns.histplot(df["actual_sales"], ax=axes[0, 0])
axes[0, 0].set_title("Raw Sales")

sns.histplot(df["log_sales"], ax=axes[0, 1])
axes[0, 1].set_title("Log Sales")

sns.histplot(df["detailing"], ax=axes[1, 0])
axes[1, 0].set_title("Raw Detailing")

sns.histplot(df["log_detailing"], ax=axes[1, 1])
axes[1, 1].set_title("Log Detailing")

sns.histplot(df["direct_to_consumer_ad"], ax=axes[2, 0])
axes[2, 0].set_title("Raw Direct-to-Consumer Ads (DTCA)")

sns.histplot(df["log_dtca"], ax=axes[2, 1])
axes[2, 1].set_title("Log Direct-to-Consumer Ads (DTCA)")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
```

```
print("After log transformation, sales and detailing distributions become much more symmetric and sprea
print("Direct-to-Consumer Ads (DTCA) also improves, but the large spike at zero remains because log(1+0
print("This spike is not a problem. It is a real feature of the data that the model will handle via the
```

```
[EDA] Comparing raw and log-transformed distributions...
```




```
[Business Insight]
After log transformation, sales and detailing distributions become much more symmetric and spread out ev
enly.
Direct-to-Consumer Ads (DTCA) also improves, but the large spike at zero remains because log(1+0) = 0 fo
r all zero-Direct-to-Consumer Ads (DTCA) observations.
This spike is not a problem. It is a real feature of the data that the model will handle via the zero-Di
rect-to-Consumer Ads (DTCA) indicator.
```

In [201…

```
print("\n[EDA] Comparing skewness before and after log transformation...")

skew_before = df[["actual_sales", "detailing", "direct_to_consumer_ad"]].skew()
skew_after = df[["log_sales", "log_detailing", "log_dtca"]].skew()

skew_compare = pd.DataFrame({
    "Before": skew_before,
    "After": skew_after
})

display(skew_compare)

print("\n[Business Insight]")
print("Skewness drops from 1.71 to -0.16 for sales, and from 1.04 to -0.61 for detailing.")
print("Values closer to zero mean the distribution is more balanced, which makes regression estimates m
```

```
[EDA] Comparing skewness before and after log transformation...
```

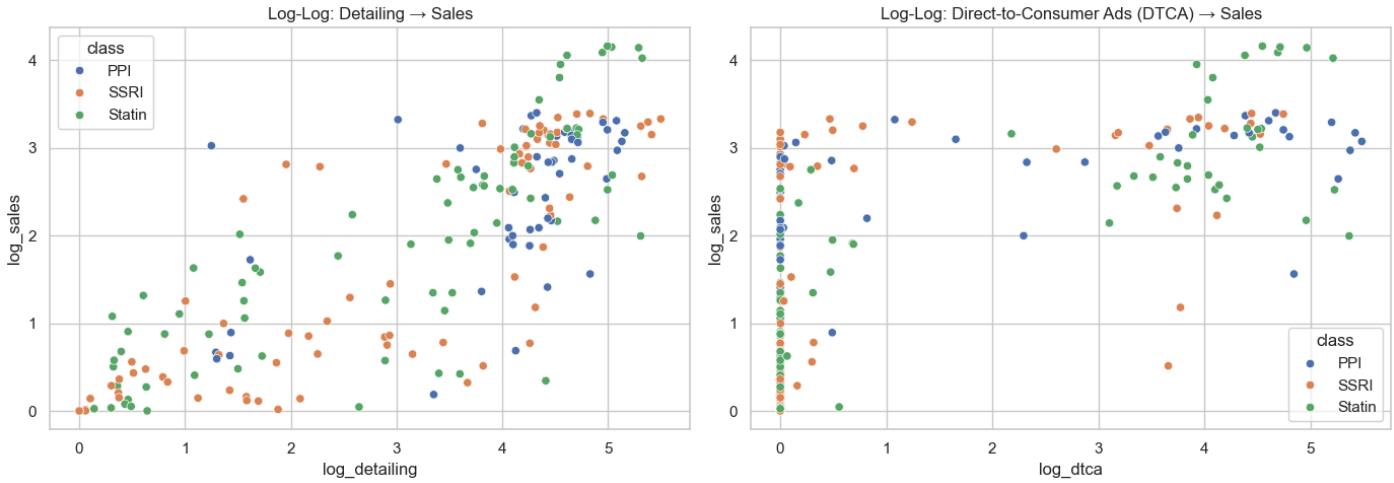
	Before	After
actual_sales	1.7127	NaN
detailing	1.0358	NaN
direct_to_consumer_ad	2.2988	NaN
log_detailing	NaN	-0.6077
log_dtca	NaN	0.8257
log_sales	NaN	-0.1557

```
[Business Insight]
Skewness drops from 1.71 to -0.16 for sales, and from 1.04 to -0.61 for detailing.
Values closer to zero mean the distribution is more balanced, which makes regression estimates more stab
le and reliable.
```

In [202…

```python
print("\n[EDA] Evaluating log-log relationships...")

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

sns.scatterplot(data=df, x="log_detailing", y="log_sales", hue="class", ax=axes[0])
axes[0].set_title("Log-Log: Detailing → Sales")

sns.scatterplot(data=df, x="log_dtca", y="log_sales", hue="class", ax=axes[1])
axes[1].set_title("Log-Log: Direct-to-Consumer Ads (DTCA) → Sales")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("In log space, the relationship between marketing spend and sales becomes much closer to a straig
print("This means a log-log regression is appropriate, where coefficients represent elasticities:")
print("e.g., a coefficient of 0.3 means a 1% increase in detailing is associated with a 0.3% increase i
```

```
[EDA] Evaluating log-log relationships...
```



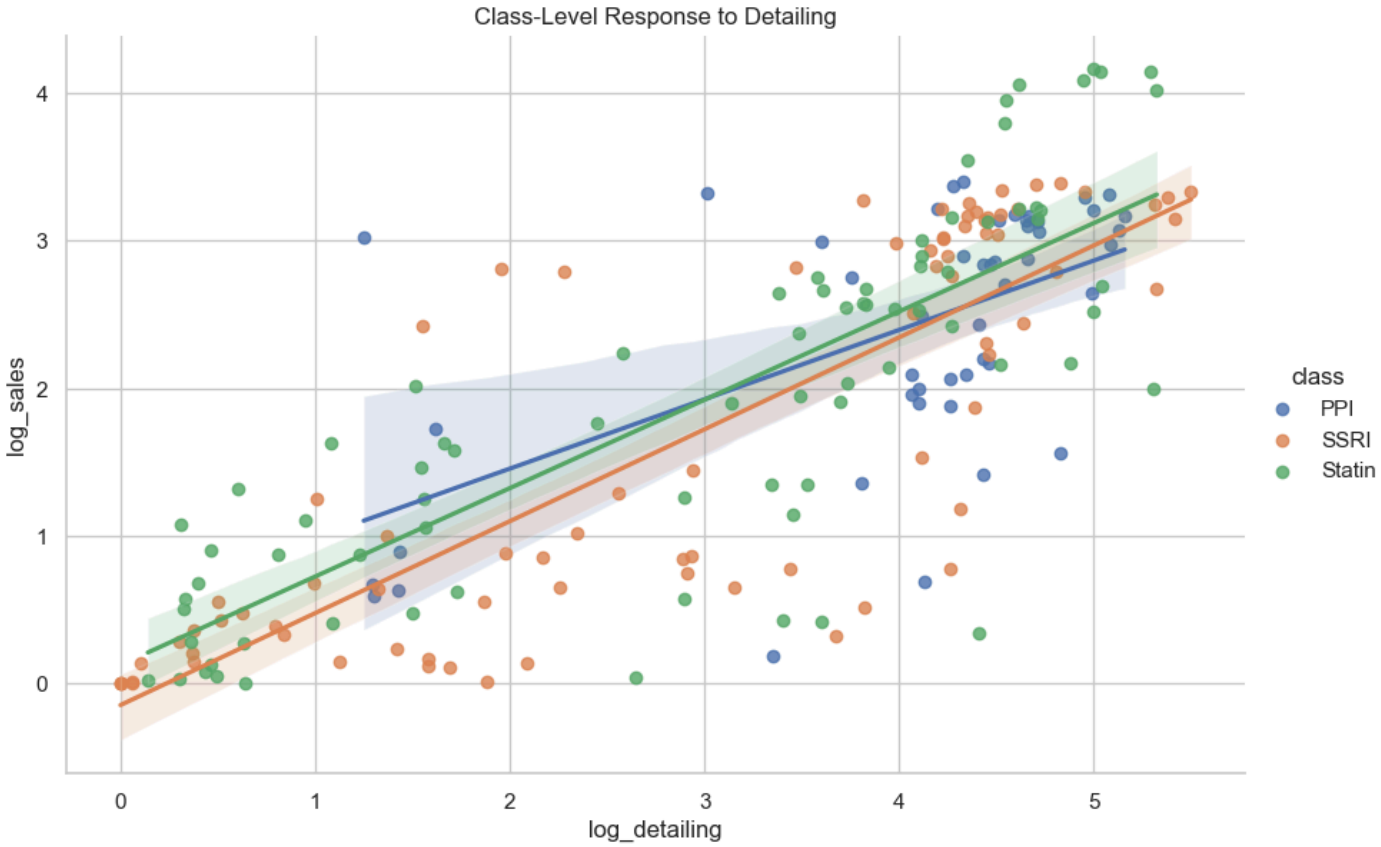

```
[Business Insight]
In log space, the relationship between marketing spend and sales becomes much closer to a straight line.
This means a log-log regression is appropriate, where coefficients represent elasticities:
e.g., a coefficient of 0.3 means a 1% increase in detailing is associated with a 0.3% increase in sales.
```

In [203…

```python
print("\n[EDA] Comparing class-level marketing sensitivity in log space...")

sns.lmplot(
    data=df,
    x="log_detailing",
    y="log_sales",
    hue="class",
    height=6,
    aspect=1.5
)

plt.title("Class-Level Response to Detailing")
plt.show()

print("\n[Business Insight]")
print("The three classes show different slopes: Statins respond more steeply to detailing than SSRIs.")
```

```
print("This means a dollar of detailing may generate different sales returns depending on the therapeut
print("The model should include class-by-detailing interaction terms to capture these differences.")
```

```
[EDA] Comparing class-level marketing sensitivity in log space...
```




```
[Business Insight]
The three classes show different slopes: Statins respond more steeply to detailing than SSRIs.
This means a dollar of detailing may generate different sales returns depending on the therapeutic clas
s.
The model should include class-by-detailing interaction terms to capture these differences.
```

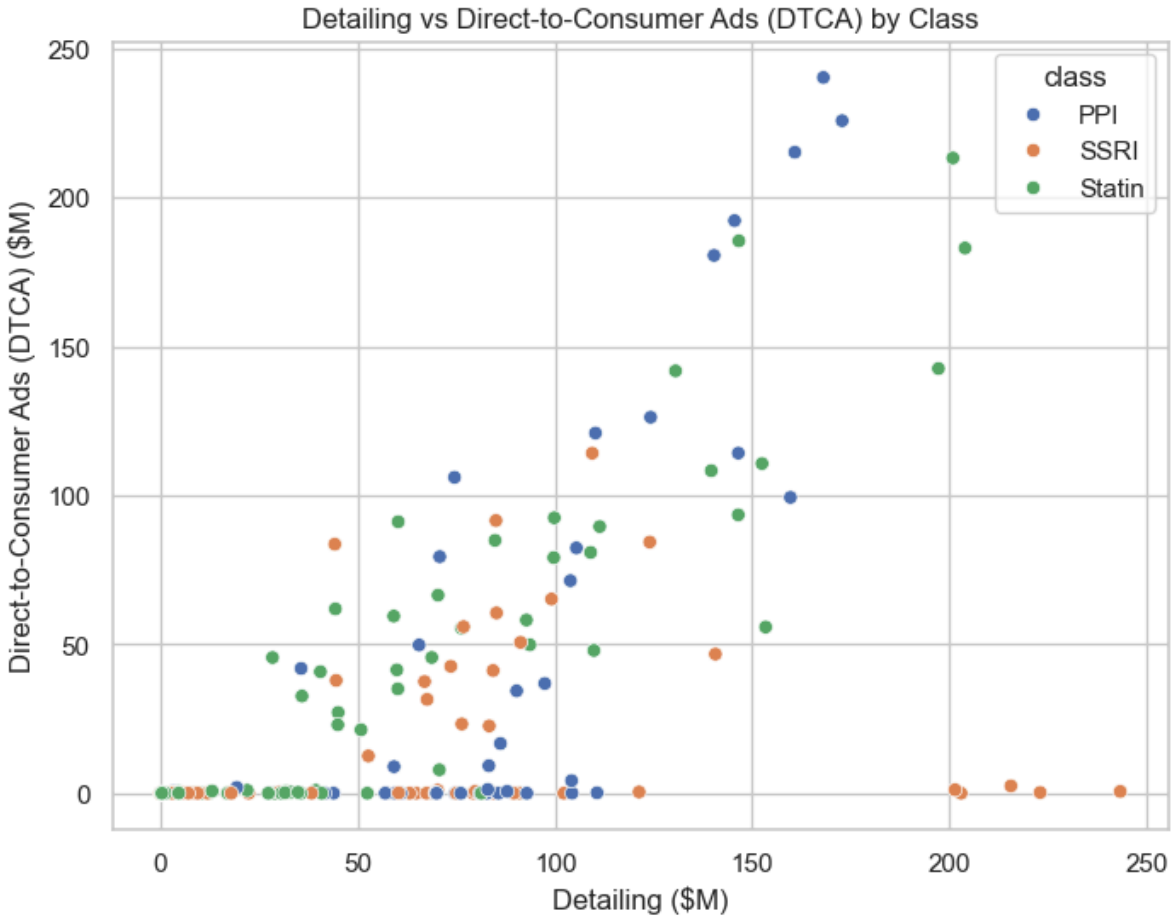
10. Cross-Channel Strategy and Collinearity

Do brands tend to invest in both channels together, or do they specialize?

In [204…

```
print("\n[EDA] Examining the relationship between detailing and Direct-to-Consumer Ads (DTCA)...")

plt.figure(figsize=(8, 6))
sns.scatterplot(
    data=df,
    x="detailing",
    y="direct_to_consumer_ad",
    hue="class"
)
plt.title("Detailing vs Direct-to-Consumer Ads (DTCA) by Class")
plt.xlabel("Detailing ($M)")
plt.ylabel("Direct-to-Consumer Ads (DTCA) ($M)")
plt.show()

print("\n[Business Insight]")
print("Some classes appear to invest jointly across both channels, while others skew more heavily towar
print("This informs both channel strategy interpretation and later multicollinearity checks.")
```

```
[EDA] Examining the relationship between detailing and Direct-to-Consumer Ads (DTCA)...
```



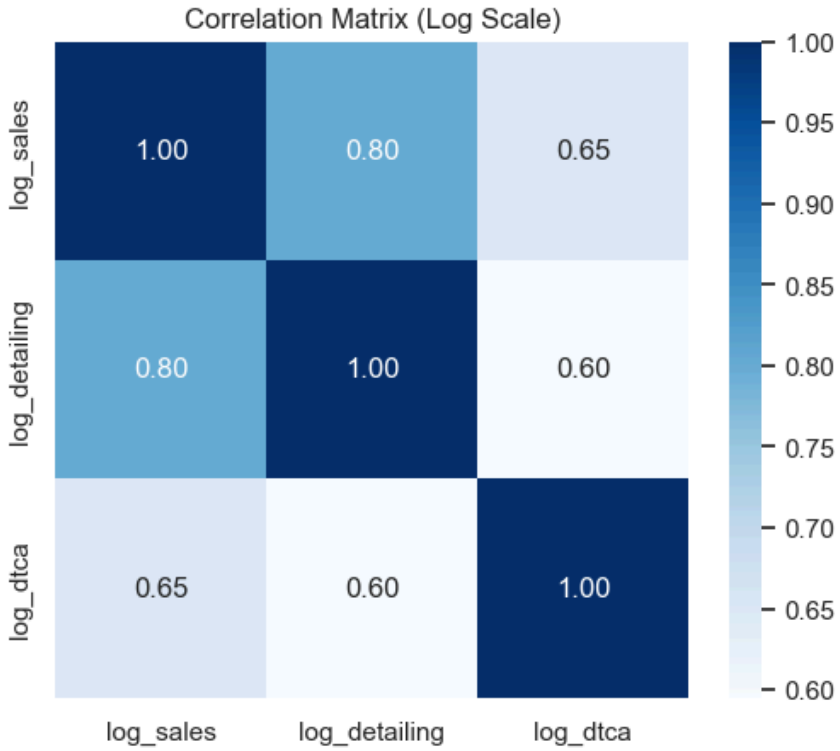

```
[Business Insight]
Some classes appear to invest jointly across both channels, while others skew more heavily toward one.
This informs both channel strategy interpretation and later multicollinearity checks.
```

In [205…

```
print("\n[EDA] Calculating correlation matrix for log-transformed variables...")

corr = df[["log_sales", "log_detailing", "log_dtca"]].corr()

display(corr)

plt.figure(figsize=(6, 5))
sns.heatmap(corr, annot=True, cmap="Blues", fmt=".2f")
plt.title("Correlation Matrix (Log Scale)")
plt.show()

print("\n[Business Insight]")
print("Detailing and Direct-to-Consumer Ads (DTCA) show moderate correlation, suggesting some coordinat
```

```
[EDA] Calculating correlation matrix for log-transformed variables...
```

	log_sales	log_detailing	log_dtca
log_sales	1.0000	0.8027	0.6482
log_detailing	0.8027	1.0000	0.5953
log_dtca	0.6482	0.5953	1.0000




```
[Business Insight]
Detailing and Direct-to-Consumer Ads (DTCA) show moderate correlation, suggesting some coordinated inves
tment strategies. While not severe, this relationship should be monitored in regression diagnostics.
```

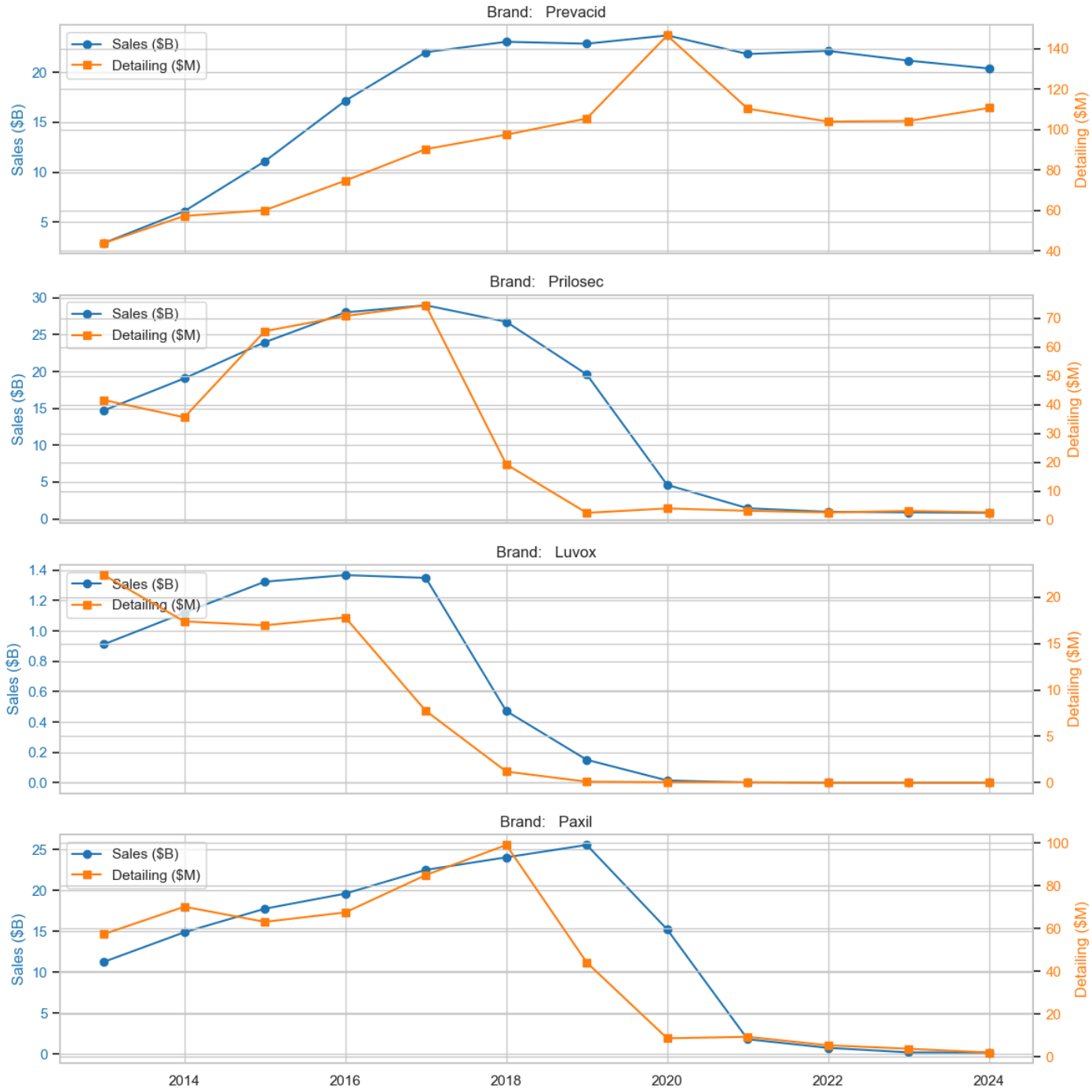
11. Panel Structure and Within-Brand Dynamics

To isolate marketing response credibly, we need to look beyond cross-sectional size and examine within-brand changes over time.

In [206…

```
print("\n[EDA] Exploring within-brand trajectories for selected brands...")

sample_agents = df["agent"].drop_duplicates().head(4).tolist()

fig, axes = plt.subplots(len(sample_agents), 1, figsize=(12, 12), sharex=True)

for i, a in enumerate(sample_agents):
    temp = df[df["agent"] == a].sort_values("year")

    color_sales = "#1f77b4"
    axes[i].plot(temp["year"], temp["actual_sales"], marker="o", color=color_sales, label="Sales ($B)")
    axes[i].set_ylabel("Sales ($B)", color=color_sales)
    axes[i].tick_params(axis="y", labelcolor=color_sales)

    ax_right = axes[i].twinx()
    color_detail = "#ff7f0e"
    ax_right.plot(temp["year"], temp["detailing"], marker="s", color=color_detail, label="Detailing ($M
    ax_right.set_ylabel("Detailing ($M)", color=color_detail)
    ax_right.tick_params(axis="y", labelcolor=color_detail)

    axes[i].set_title(f"Brand: {a}")
    lines_1, labels_1 = axes[i].get_legend_handles_labels()
    lines_2, labels_2 = ax_right.get_legend_handles_labels()
    axes[i].legend(lines_1 + lines_2, labels_1 + labels_2, loc="upper left")

plt.tight_layout()
plt.show()

print("\n[Business Insight]")
print("When a single agent (brand) increases its detailing spend over time, its sales tend to move in t
print("This within-brand co-movement is important because it helps the model separate the effect of mar
print("from the effect of simply being a bigger or smaller agent (brand).")
```

```
[EDA] Exploring within-brand trajectories for selected brands...
```




```
[Business Insight]
When a single agent (brand) increases its detailing spend over time, its sales tend to move in the same
direction.
This within-brand co-movement is important because it helps the model separate the effect of marketing
from the effect of simply being a bigger or smaller agent (brand).
```

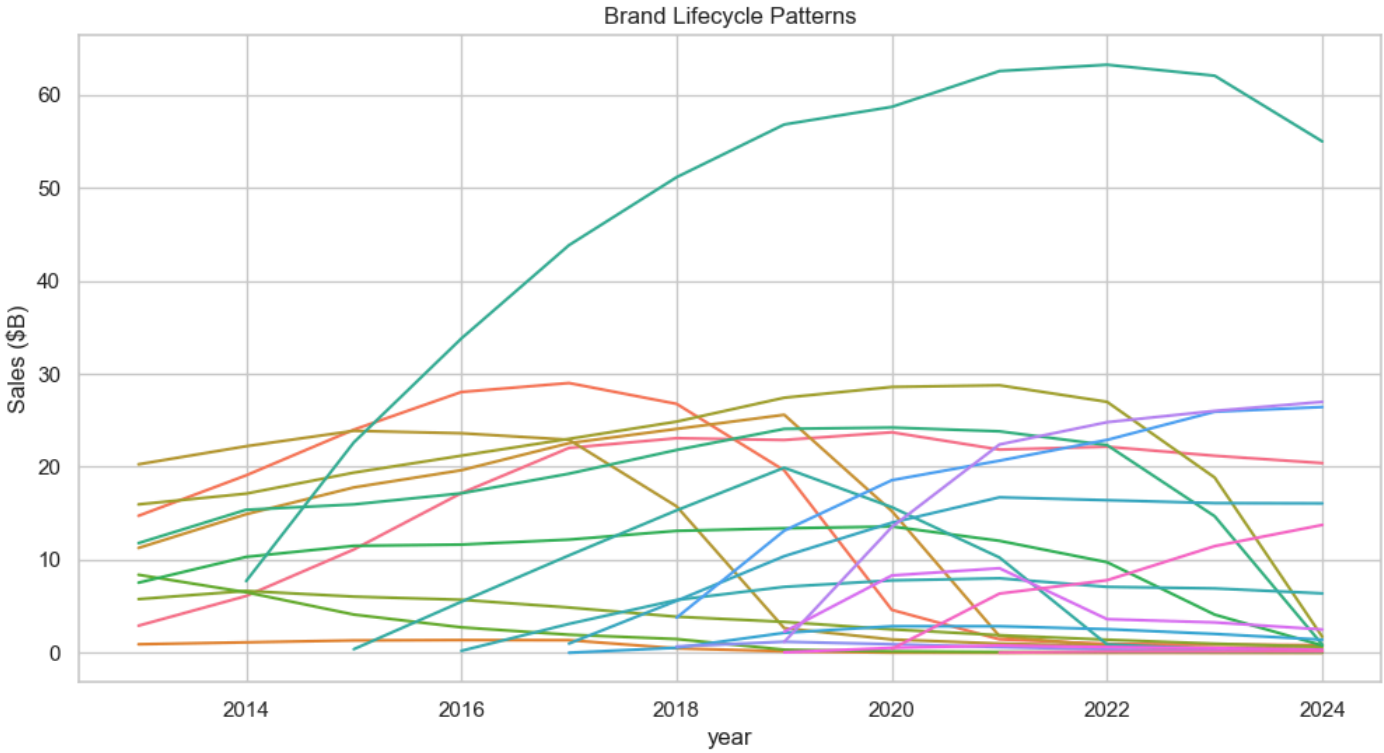
In [207…

```python
print("\n[EDA] Inferring brand lifecycle patterns...")

agent_lifecycle = (
    df.groupby("agent")
    .agg(
        start_year=("year", "min"),
        end_year=("year", "max"),
        avg_sales=("actual_sales", "mean")
    )
    .sort_values("start_year")
)

display(agent_lifecycle.head(15))

plt.figure(figsize=(12, 6))
sns.lineplot(data=df, x="year", y="actual_sales", hue="agent", legend=False)
plt.title("Brand Lifecycle Patterns")
plt.ylabel("Sales ($B)")
plt.show()

print("\n[Business Insight]")
print("Some agents (brands) like Lipitor show strong growth followed by a plateau, while others like Lu
```

```
print("A agent (brand) in its growth phase may show high marketing returns simply because it is growing
print("The model needs to account for these lifecycle differences to avoid crediting marketing for grow
```

```
[EDA] Inferring brand lifecycle patterns...
```

agent	start_year	end_year	avg_sales
Zoloft	2013	2024	21.1441
Prozac	2013	2024	11.2791
Prilosec	2013	2024	14.1557
Lescol	2013	2024	3.6356
Prevacid	2013	2024	17.8747
Pravachol	2013	2024	9.9861
Luvox	2013	2024	0.5595
Mevacor	2013	2024	2.1499
Zocor	2013	2024	17.6050
Paxil	2013	2024	12.8184
Lipitor	2014	2024	47.0275
Celexa	2015	2024	7.8978
Aciphex	2016	2024	5.8032
Lescol XL	2017	2024	1.7925
Protonix	2017	2024	12.0249




```
[Business Insight]
Some agents (brands) like Lipitor show strong growth followed by a plateau, while others like Luvox decl
ine steadily.
A agent (brand) in its growth phase may show high marketing returns simply because it is growing, not be
cause marketing is more effective.
The model needs to account for these lifecycle differences to avoid crediting marketing for growth that
would have happened anyway.
```

12. Outliers and High-Leverage Observations

Before modeling, we identify whether a few brands or campaigns are likely to dominate results.

In [208…

```
print("\n[EDA] Reviewing highest-leverage observations...")

print("Top 5 by sales:")
```

```
display(df.nlargest(5, "actual_sales")[["class", "agent", "year", "actual_sales", "detailing", "direct_

print("Top 5 by detailing:")
display(df.nlargest(5, "detailing")[["class", "agent", "year", "actual_sales", "detailing", "direct_to_

print("Top 5 by Direct-to-Consumer Ads (DTCA):")
display(df.nlargest(5, "direct_to_consumer_ad")[["class", "agent", "year", "actual_sales", "detailing",

print("\n[Business Insight]")
print("Lipitor dominates the top 5 by sales, Lexapro dominates the top 5 by detailing, and Nexium domin
print("These are not data errors. They are real, high-value agents (brands) that naturally stand out.")
print("However, because they are so extreme, they could disproportionately influence model results.")
print("During modeling, we will run robustness checks by excluding these agents (brands) to see if the
```

```
[EDA] Reviewing highest-leverage observations...
Top 5 by sales:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
159	Statin	Lipitor	2022	63.2186	146.4874	93.4800
137	Statin	Lipitor	2021	62.5406	152.5107	110.7082
181	Statin	Lipitor	2023	62.0389	197.2503	142.6181
115	Statin	Lipitor	2020	58.6918	139.6403	108.3209
94	Statin	Lipitor	2019	56.7952	99.7099	79.1343

```
Top 5 by detailing:
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
191	SSRI	Lexapro	2024	26.9766	243.3303	0.5963
125	SSRI	Lexapro	2021	22.3924	223.0231	0.2595
169	SSRI	Lexapro	2023	26.0019	215.6069	2.4667
203	Statin	Lipitor	2024	54.9727	203.9765	183.0804
104	SSRI	Lexapro	2020	13.5176	202.9746	0.0000

```
Top 5 by Direct-to-Consumer Ads (DTCA):
```

	class	agent	year	actual_sales	detailing	direct_to_consumer_ad
120	PPI	Nexium	2021	20.6499	168.0683	240.3550
142	PPI	Nexium	2022	22.8833	172.8186	225.8439
99	PPI	Nexium	2020	18.5523	160.8279	215.2616
134	Statin	Crestor	2021	6.3594	200.9031	213.3158
79	PPI	Nexium	2019	13.1240	145.5416	192.2922

```
[Business Insight]
Lipitor dominates the top 5 by sales, Lexapro dominates the top 5 by detailing, and Nexium dominates the
top 5 by DTCA.
These are not data errors. They are real, high-value agents (brands) that naturally stand out.
However, because they are so extreme, they could disproportionately influence model results.
During modeling, we will run robustness checks by excluding these agents (brands) to see if the conclusi
ons change.
```

13. EDA Takeaways

In [209…

```
print("\n[EDA] Summarizing final business takeaways...")

print("""
1. The portfolio is highly concentrated, with a small number of blockbuster brands driving over half of
2. Sales exhibit a clear lifecycle pattern, rising through mid-period before declining, indicating that
3. Marketing investments are not uniformly allocated — both channel mix and spend levels vary significa
4. Simple efficiency ratios suggest differences across channels, but are unstable due to structural zer
5. There is consistent evidence of diminishing returns, particularly in detailing, supporting nonlinear
6. Both class and brand identity materially influence baseline sales and likely response to marketing,
7. Nearly half of observations have zero Direct-to-Consumer Ads (DTCA), indicating a distinct physician
8. The dataset is an unbalanced panel, requiring careful time-based validation to avoid bias from brand
9. Log transformations substantially improve distributional properties and linearity, enabling elastici
10. Overall, the evidence supports a log-log parametric model with class, brand, and time controls, and
""")
```

```
[EDA] Summarizing final business takeaways...

1. The portfolio is highly concentrated, with a small number of blockbuster brands driving over half of
total revenue.
2. Sales exhibit a clear lifecycle pattern, rising through mid-period before declining, indicating that
time dynamics must be explicitly modeled.
3. Marketing investments are not uniformly allocated — both channel mix and spend levels vary significan
tly across classes and brands.
4. Simple efficiency ratios suggest differences across channels, but are unstable due to structural zero
s and should not be interpreted causally.
5. There is consistent evidence of diminishing returns, particularly in detailing, supporting nonlinear
or log-transformed specifications.
6. Both class and brand identity materially influence baseline sales and likely response to marketing, j
ustifying their inclusion as structural controls.
7. Nearly half of observations have zero Direct-to-Consumer Ads (DTCA), indicating a distinct physician-
driven marketing regime that must be modeled separately.
8. The dataset is an unbalanced panel, requiring careful time-based validation to avoid bias from brand
entry and exit.
9. Log transformations substantially improve distributional properties and linearity, enabling elasticit
y-based interpretation of model coefficients.
10. Overall, the evidence supports a log-log parametric model with class, brand, and time controls, and
a potential regime distinction for Direct-to-Consumer Ads (DTCA) activity.
```

Next Steps for Parametric Modeling

This EDA has already prepared the key features needed for the next stage:

- `log_sales`
- `log_detailing`
- `log_dtca`
- `dtca_zero`

Recommended first parametric baseline

[\log(1 + Sales) \sim \log(1 + Detailing) + \log(1 + DTCA) + Class + Agent + Year]

Extensions suggested by the EDA

- class × channel interaction terms
- `dtca_zero` as a binary regime indicator
- nonlinear response checks if residual diagnostics suggest misfit
- robustness checks for high-leverage brands

In [210…

```
print("\n[EDA] Saving processed dataset for the next modeling stage...")

df.to_csv("data/processed_data.csv", index=False)

print("[EDA] Saved processed file: data/processed_data.csv")
```

```
[EDA] Saving processed dataset for the next modeling stage...
[EDA] Saved processed file: data/processed_data.csv
```

In [210…